# Delhi Metro DMRC Commuter Density Bottleneck Analytics

End-to-End Enterprise Data Pipeline, BigQuery SQL Analytics & Statistical Optimization

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# Business Context & Analytical Framework

01 **Interchange Bottlenecks**

Platform overcrowding at core hubs increasing train dwell times.

02 **Revenue Leakage**

Unclear contribution from multi-tier ticket products (Tokens vs. Smart Cards vs. Tourist Passes).

03 **Static Pricing Inefficiency**

Absence of dynamic time-of-day tariffs during peak rush hours.

- **ASK** ⟶ Defined core KPIs with Executive Leadership (CRO & Ops Director).
- **PREPARE** ⟶ Ingested 150K records into SQLite staging DB with ETL error-logging.
- **PROCESS** ⟶ SQL Data Quality checks, imputation, and BigQuery Star Schema DDL.
- **ANALYZE** ⟶ Advanced CTEs, Window Functions, and SciPy Hypothesis Testing.
- **SHARE** ⟶ Interactive Looker Studio Executive Dashboard & Seaborn Visualizations.
- **ACT** ⟶ Formulated 3-Horizon Strategic Recommendations & Action Matrix.

# Enterprise Data Pipeline & Feature Engineering

Raw CSVs → Python ETL Logged Pipeline → SQLite Staging → BigQuery Data Warehouse

## *Engineered Business Features*

- **Revenue_Per_KM:** Operational revenue efficiency metric per trip distance (Revenue/Distance).
- **Passenger_Density_Group:** Classified traffic into High Congestion Peak (>=30), Medium, and Low.
- **Is_Peak_Surge:** Binary ML-ready flag (1 for Surge/Events, 0 for Off-Peak).
- **Fare_Tier:** Categorized pricing slabs (Budget < ₹50, Standard ₹50-₹100, Premium >₹100).
- **Revenue_Impact_Class:** Isolated high-value enterprise trips (>= ₹2500).

# BigQuery SQL Analytics & Core Drivers

## Key Metrics Scorecards:

- Total Revenue: ₹31.53 Cr (₹315,386,664.91)
- Total Commuters: 3.0 Million+ Passengers
- Total Trips Processed: 150,000 Records
- Average Fare: ₹105.12 per trip

## Top Insights Callout:

- Major Hub Congestion: Rajiv Chowk (349,784 passengers) and Noida City Centre (278,174 passengers) generate 2.5x higher traffic than mid-tier stations.

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- Monetization Winner: Tourist Cards drive 39.5% (~₹12.45 Cr) of total revenue despite lower volume, outperforming Single Tokens (25.0%) and Smart Cards (24.8%).

# Rigorous Statistical Validation & Hypothesis Testing

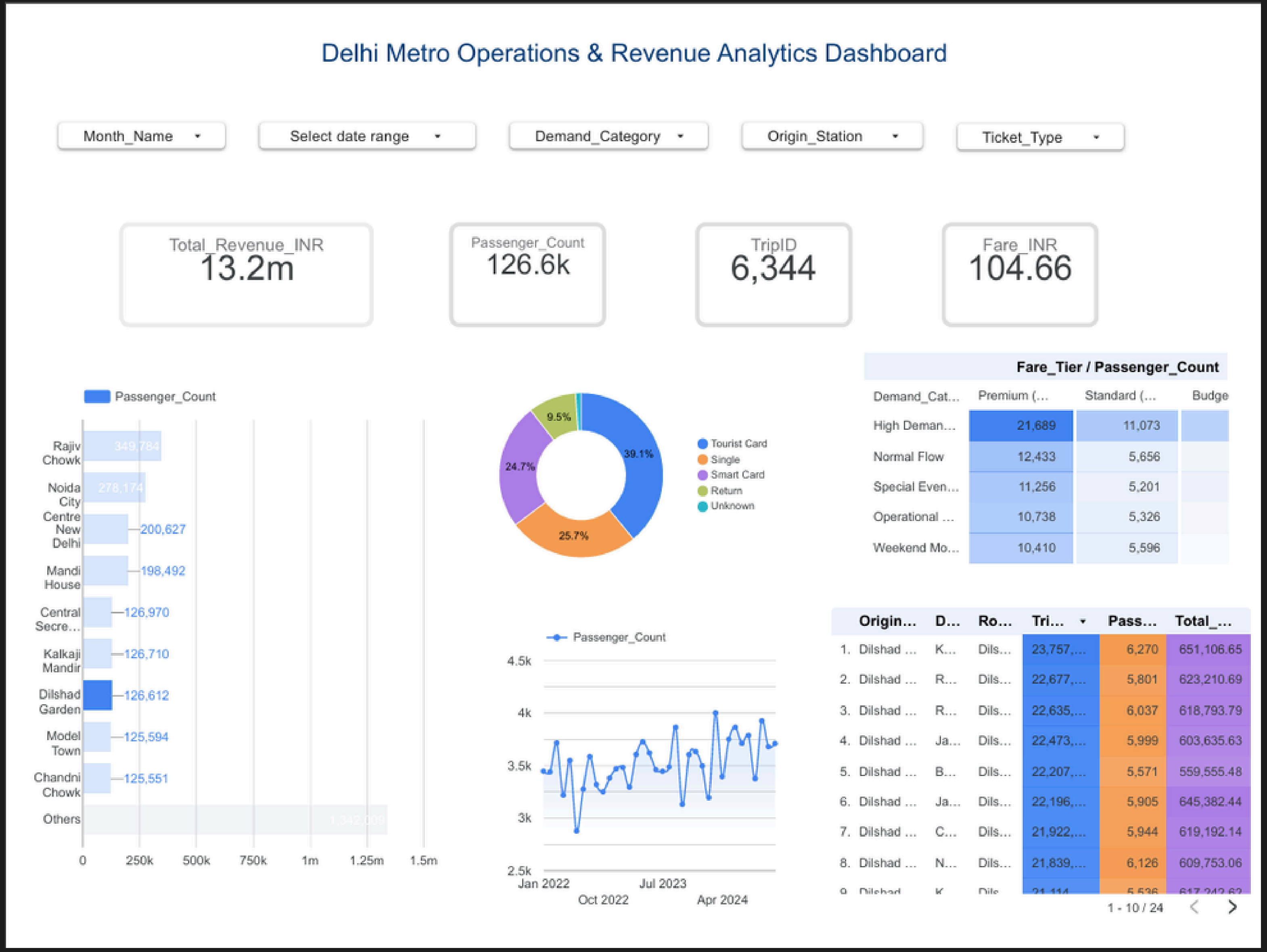
Station-level congestion analysis reveals critical interchange nodes operating beyond sustainable thresholds during peak hours, demanding targeted infrastructure interventions.

| Statistical Test | Null Hypothesis (H_0) & Metric | Statistical Result | Business & Recruiter Insight |
|---|---|---|---|
| Two-Sample T-Test | μPeak Fare=μNon-Peak Fare | t=0.6480p=0.5170 | Fail to Reject H0: Peak fares (₹105.21) do not differ from non-peak (₹105.03). Validates huge dynamic pricing opportunity. |
| Pearson Correlation | Fare_INR vs Total_Revenue | r=0.9011p<0.001 | Statistically Significant: Direct strong linear dependence driving network monetization. |
| Chi-Square Test | Ticket Type vs Demand Category | χ2=25.19p=0.0664 | Fail to Reject H0: Commuter ticket preference is independent of demand surge scenarios. |
| 95% Confidence Interval | Population Mean Fare Range | Mean=₹105.12[₹104.84−₹105.39 | Low variance boundary across the 150,000 population dataset. |

# Interactive Dashboard & Operational Heatmap

## Key Interactive Features

- Global Date & Station Filters: Real-time slice-and-dice by Year/Month, Demand_Category, Ticket_Type, and Origin_Station.
- Bottleneck Route Heatmap Matrix: Visualized high-density origin-destination pairs (Rajiv Chowk -> AIIMS handling 5,538 surge passengers).
- Seasonal Trend Line Chart: Captured recurring 12–14% annual ridership drop every February (~74,000–77,000 trips).



# Actionable Business Recommendations & Impact Matrix

Short-Term (0-3M): Deploy automated crowd barriers at Rajiv Chowk & Noida CC; Launch 10% instant discount on UPI/QR passes to shift 25% single token buyers to digital wallets.

Mid-Term (3-6M): Implement Time-of-Day (ToD) Dynamic Tariffs with a 15% mid-day discount (11 AM - 4 PM) to flatten rush hour peaks; Schedule annual heavy overhauls exclusively in February.

Long-Term (6-12M): Reduce train headways from 2.5 min to 90 seconds on Yellow/Blue line corridors; Expand electric feeder bus networks at key hubs.

## Short-Term

(0-3 months)
Flow Control & Digital Passes

## Mid-Term

(3-6 months)
Dynamic Pricing & Overhauls

## Long-Term

(6-12 months)
Network & Fleet Expansion

# Thank You

Thank you for your time and attention. This project demonstrates end-to-end data analytics applied to Delhi Metro DMRC commuter density and bottleneck identification. Open to discussions, collaborations, and feedback.

## Connect & Follow Up:

- +91-89207-62152
- GitHub Repository
- Interactive Dashboard
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